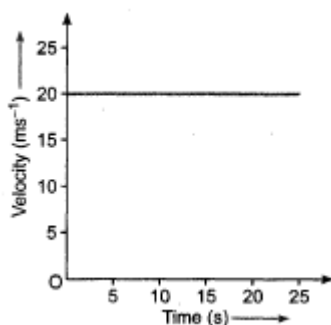




MOTION-03

Class 09 - Science

1. A body travels along a circular path of radius 70m. After travelling half a revolution in 20 s, find the (i) average velocity (ii) average speed, [2]
2. What happens when you shake a wet piece of cloth? Explain your observation. [2]
3. Mention some uses of velocity time graphs. [2]
4. Obtain a relation for the distance travelled by an object moving with a uniform acceleration in the interval between 4th and 5th second. [2]
5. What is the nature of the distance time graphs for uniform and non-uniform motion of an object? [2]
6. A train is travelling at a speed of 90 kmh⁻¹. Brakes are applied so as to produce a uniform acceleration of -0.5 ms⁻². Find how far the train will go before it is brought to rest. [2]
7. A stone is thrown in a vertically upward direction with a velocity of 5 ms⁻¹. If the acceleration of the stone during its motion is 10 ms⁻² in the downward direction, what will be the height attained by the stone and how much time will it take to reach there? [2]
8. Define uniform velocity and uniform acceleration. [2]
9. A racing car has uniform acceleration of 4 ms⁻². What distance will it cover in 10 s after start? [2]
10. An electric train is moving with a velocity of 120kmh⁻¹. How much distance will it move in 30s? [2]
11. The brakes applied to a car produce an acceleration of 6 ms⁻² in the opposite direction to the motion. If the car takes 2s to stop after the application of brakes, calculate the distance it travels during this time. [2]
12. During an experiment, a signal from a spaceship reached the ground station in five minutes. What was the distance of the spaceship from the ground station? The signal travels at the speed of light, that is 3×10^8 ms⁻¹. [2]
13. The velocity-time graph shows the motion of a cyclist. Find [2]
 - i. its acceleration
 - ii. its velocity and
 - iii. the distance covered by the cyclist in 15 seconds.



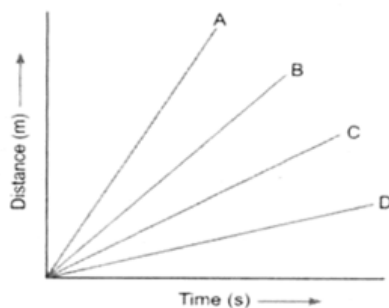
14. The distance between two stations is 200 km. A train travels for the first 100 km at a speed of 10 kmh⁻¹. How [2]

fast should the train travel the next 100 km so as to average 70 kmh^{-1} for the whole journey?

15. Distinguish between speed and velocity. [2]
16. Under what condition(s) is the magnitude of average velocity of an object equal to its average speed? [2]
17. What can you say about the motion of an object whose distance-time graph is a straight line parallel to the time axis? [2]
18. A trolley, while going down an inclined plane, has an acceleration of 2 cm/s^2 . What will be its velocity after 3 seconds? [2]
19. A car acquires a velocity of 72 kmh^{-1} in 10 second after starting from rest. Find (a) the acceleration (b) the average velocity and (c) the distance travelled in this time. [2]
20. A ball is gently dropped from a height of 20 m. If its velocity increases uniformly at the rate of 10 ms^{-2} , with what velocity will it strike the ground? After what time will it strike the ground? [2]
21. A train starting from rest attains a velocity of 72 km h^{-1} in 5 minutes. Assuming that the acceleration is uniform, find [2]
 - i. the acceleration and
 - ii. the distance travelled by train for attaining this velocity.
22. A motorcyclist drives from A to B with a uniform speed of 30 km h^{-1} and returns back with a speed of 20 km h^{-1} . Find its average speed. [2]
23. With the help of a graph, derive the relation $v = u + at$. [2]
24. Under what conditions is the magnitude of average velocity of an object equal to its average speed? [2]
25. What are the characteristics of distance-time graph for an object moving with a non-uniform speed? [2]
26. Find the initial velocity of a car if it can be stopped in 10 sec by applying brakes which provide it a retardation of 2.5 ms^{-2} . [2]
27. A car travels a certain distance with a speed of 40 kmh^{-1} and returns with a speed 40 kmh^{-1} . Calculate the average speed for the entire journey? [2]
28. Usha swims in a 90 m long pool. She covers 180 m in one minute by swimming from one end to the other and back along the same straight path. Find the average speed and average velocity of Usha. [2]
29. An object has moved through a distance. Can it have zero displacement? If yes, support your answer with an example. [2]
30. An object has moved through a distance. Can it have zero displacement? If yes, explain with an example. [2]
31. Express average velocity when the velocity of a body changes at non-uniform rate and at uniform rate. [2]
32. Abdul while driving to school computes the average speed for his trip to be 20 kmh^{-1} . On his return trip along the same route, there is less traffic and the average speed is 40 kmh^{-1} . What is the average speed for Abdul's trip? [2]
33. A bus decreases its speed from 80 kmh^{-1} to 60 km h^{-1} in 5 s. Find the acceleration of the bus. [2]
34. When will you say a body is in [2]
 - i. uniform acceleration.
 - ii. non-uniform acceleration?
35. What do you understand by a non-uniform velocity? [2]
36. A body travels along a circular path of radius 70 m. After travelling half a revolution in 20 s, find the [2]
 - (i) average velocity,

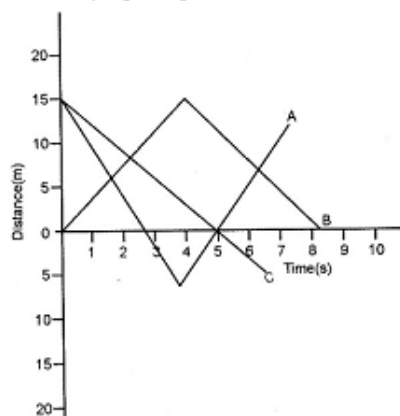
(ii) average speed.

37. Distinguish between terms speed and velocity. [2]
38. The driver of a train travelling at 40 ms^{-1} applies the brakes as a train enters a station. The train slows down at a rate of 2 ms^{-2} . The platform is 400 m long. Will the train stop in time? [2]
39. A bus decreases its speed from 80 km h^{-1} to 60 km h^{-1} in 5 s. Find the acceleration of the bus. [2]
40. A train starting from a railway station and moving with uniform acceleration attains a speed 40 kmh^{-1} in 10 minutes. Find its acceleration. [2]
41. An artificial satellite is moving in a circular orbit of radius nearly 42,250 km. Calculate its linear velocity, if it takes 24 hour to revolve round the earth. [2]
42. The odometer of a car reads 2000 km at the start of a trip and 2400 km at the end of the trip. If the trip took 8 h, calculate the average speed of the car in kmh^{-1} and ms^{-1} [2]
43. A car starts from rest and moves along the X-axis with constant acceleration 5 ms^{-2} for 8 seconds. If it is continue with constant velocity, what distance will the car cover in 12 seconds since it started from the rest? [2]
44. How will the equations of motion for an object moving with a uniform velocity change? [2]
45. An athlete completes one round of a circular track of diameter 200 m in 40 s. What will be the distance covered and the displacement at the end of 2 minutes 20 s? [2]
46. What is the difference between uniform velocity and non-uniform velocity? [2]
47. A bus starting from rest moves with a uniform acceleration of 0.1 ms^{-2} for 2 minutes. Find a) the speed acquired b) the distance travelled. [2]
48. Four cars A, B, C and D are moving on a leveled road. Their distance versus time graphs are shown in figure. Which car is the slowest? [2]

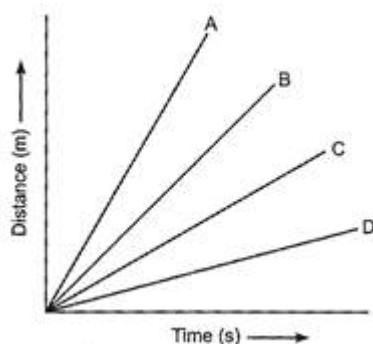


49. Draw a velocity versus time graph of a stone thrown vertically upward and then coming downwards after attaining the maximum height. [2]
50. An object is dropped from rest at a height of 150 m and simultaneously another object is dropped from rest at a height 100 m. What is the difference in their heights after 2 s if both the objects drop with same accelerations. How does the difference in heights vary with time? [2]
51. Distinguish between terms distance and displacement. [2]
52. A body is moving with a uniform velocity of 10 ms^{-1} . Find its velocity after 10 s? [2]
53. Derive the second equation of motion, $s = ut + \frac{1}{2}at^2$ numerically. [2]
54. A farmer moves along the boundary of a square field of side 10 m. in 40 s. What will be the magnitude of displacement of the farmer at the end of 2 minutes 20 seconds from his initial position [2]
55. A bus decreases its speed from 80 kmh^{-1} to 60 kmh^{-1} in 5 s. Find the acceleration of the bus. [2]
56. A body starts to slide over a horizontal surface with an initial velocity of 0.5 m/s. Due to friction, its velocity decreases at the rate of 0.05 m/s^2 . How much time will it take for the body to stop? [2]

57. Abdul, while driving to school, computes the speed for his trip to be 20 km/h. On his return trip along the same route, there is less traffic and the average speed is 30 km/h. What is the average speed for Abdul's trip? [2]
58. An artificial satellite is moving in a circular orbit of radius 42250 km. Calculate its speed if it takes 24 hours to revolve around the earth. [2]
59. An electric engine has a velocity of 120 kmh^{-1} . How much distance will it travel in 30 s? [2]
60. A motorboat starting from rest on a lake accelerates in a straight line at a constant rate of 3.0 ms^{-2} for 8.0 s. How far does the boat travel during this time? [2]
61. Differentiate between distance and displacement? [2]
62. A farmer moves along the boundary of a square field of side 10 m in 40 s. What will be the magnitude of displacement of the farmer at the end of 2 minutes and 20 seconds? [2]
63. The velocity of a car is 18 ms^{-1} . Express this velocity in kmh^{-1} . [2]
64. A trolley, while going down an inclined plane, has an acceleration of 2 cms^{-2} . What will be its velocity 3s after the start? [2]
65. A stone is thrown in vertically upward direction with a velocity of 5 ms^{-1} . If the acceleration of the stone during its motion be 10 ms^{-2} in the downward direction what will be the height attained by the stone and how much time will it take to reach there? [2]
66. The displacement of a moving object in a given interval of time is zero. Would the distance traveled by the object also be zero? Justify your answer. [2]
67. Discuss the graphs A, B and C shown in the figure. Compare the total distance travelled and the displacements. [3]
Which graph represents a motion with negative acceleration?



68. Four cars A, B, C and D are moving on a leveled road. Their distance versus time graphs are shown in figure. [3]
Which car is the slowest.



69. A body is dropped from a height of 320 m. The acceleration due to the gravity is 10 m/s^2 . [3]
(a) How long does it take to reach the ground?
(b) What is the velocity with which it will strike the ground?

70. A bus starting from rest moves with a uniform acceleration of 0.1 m s^{-2} for 2 minutes. Find: [3]
 a. the speed acquired.
 b. the distance travelled.
71. An electron moving with a velocity of $5 \times 10^4 \text{ ms}^{-1}$ enters into a uniform electric field and acquires a uniform acceleration of 10^4 ms^{-2} in the direction of its initial motion. [3]
 i. Calculate the time in which the electron would acquire a velocity double of its initial velocity.
 ii. How much distance the electron would cover in this time?
72. A cheetah is the fastest land animal and can achieve a peak velocity of 100 kmh^{-1} up to distances less than 500 m . If a cheetah spots his prey at a distance of 100 m, what is the minimum time it will take to get its prey, if the average velocity attained by it is 90 kmh^{-1} ? [3]
73. A train is travelling at a speed of 90 kmh^{-1} . Brakes are applied so as to produce a uniform retardation of 0.5 ms^{-2} . Find how far the train will go before it is brought to rest. [3]
74. A particle moves in a circle with O as centre and $AO = OB = 5 \text{ cm}$, radius, as shown in the figure. It starts from A. Calculate. [3]



- a. the distance covered, and
 b. the displacement, when it reaches B.
75. Suppose you go up a tower 80 m high and throw a ball horizontally with a velocity of 20m/s. What will be the shape of the path followed by the ball? While falling, the motion of the ball will be a combination of two independent motions. Name these two motions. [3]